Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Data Visualization with Power BI and Tableau

TABLEAU DASHBOARD PROJECT

DMart Sales, Profitability & Customer Segmentation Dashboard

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Class / Batch	TYIT
Subject	Data Visualization with Power BI and Tableau
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1. Project Title

DMart Sales, Profitability & Customer Segmentation Dashboard using Tableau

2. Objective

The main objective of this project is to create an interactive Tableau dashboard for analyzing DMart sales, profit, regional performance, discount impact, product/brand performance, and customer behavior. The dashboard converts transaction data into meaningful visual information and helps users understand sales trends, profitability, customer segments, and high-value customers.

3. Project Description

This project uses Tableau to develop an interactive DMart Sales, Profitability & Customer Segmentation Dashboard. The dashboard contains KPI cards, charts, a geographical state view, filters, RFM analysis, clustering, and high-value customer analysis. Users can analyze sales and profit by brand, state, region, category, discount, and time period. RFM (Recency, Frequency, Monetary Value) analysis and Tableau clustering are used to identify groups of customers with different purchasing patterns.

4. Dataset Details

Parameter	Details
Dataset Name	DMart Tableau – 5,000 Transactions
Source	Custom-generated synthetic dataset created from the DMart product/catalog data for academic Tableau analysis
Number of Records	5,000
Important Fields	Sale_ID, Sale_Date, Customer_ID, Customer_Name, Name, Brand, Category, SubCategory, Quantity, Selling_Price, Discount, DiscountedPrice, Sales, Profit, Region, State, City, Payment_Method

5. Tools & Technologies Used

- Tableau Desktop / Tableau Public
- Microsoft Excel / CSV for dataset preparation
- Data visualization and dashboarding

6. Tableau Worksheets / Charts

Visualization	Purpose
KPI Cards	Display Total Sales, Total Profit, Products Sold, Profit Ratio and Total Customers
Line Chart	Show monthly sales trends from 2022–2025

Bar Chart	Compare sales performance across DMart brand values
Map	Visualize sales performance across Indian states
Scatter Plot	Analyze the relationship between discount and profit
RFM Clustering	Segment customers using Recency, Frequency and Monetary Value with Tableau clustering
High-Value Customer Analysis	Identify customers with high monetary value
Interactive Filters	Filter the dashboard by Region, Brand, Category and Year
Dashboard Actions	Allow map and brand selections to filter other dashboard views

7. Calculated Fields

The following calculated fields were used in the Tableau dashboard:

Total Sales:

```
SUM([Sales])
```

Total Profit:

```
SUM([Profit])
```

Profit Ratio:

```
SUM([Profit]) / SUM([Sales])
```

Frequency:

```
COUNTD([Sale_ID])
```

Monetary Value:

```
SUM([Sales])
```

Recency:

```
DATEDIFF('day', { FIXED [Customer_ID] : MAX([Sale_Date]) }, { MAX([Sale_Date]) })
```

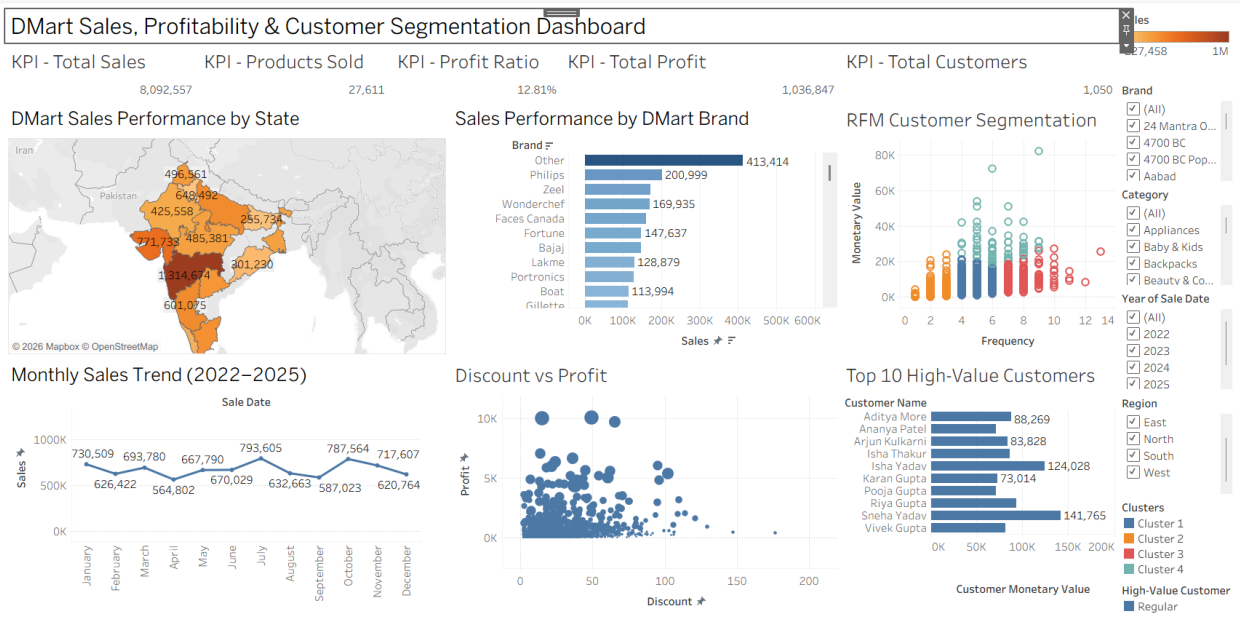
Customer Monetary Value:

```
{ FIXED [Customer_ID] : SUM([Sales]) }
```

High-Value Customer:

```
IF [Customer Monetary Value] >= 40000
THEN "High-Value"
ELSE "Regular" END
```

8. Dashboard Screenshot



9. Key Insights

1. **Overall Performance:** The dataset contains 5,000 transactions, generating ₹8,092,557.03 in sales and ₹1,036,846.70 in profit, with a profit ratio of 12.81%.
2. **Customer Base:** The dashboard covers 1,050 unique customers, allowing customer-level segmentation and high-value customer analysis.
3. **Regional** Performance: The West region contributes the highest sales at ₹2,997,346.06; East is the lowest among the four regions in this dataset.
4. **Category Performance:** Home & Kitchen is the leading category with ₹1,984,862.41 in sales.
5. **High-Value Customers:** Using the dataset-adapted ₹40,000 threshold, 9 customers qualify as high-value.

10. Video Demonstration

Video Link: https://youtu.be/h3jJ1LO1JWs?si=mM83HS2djrWP4ORH

11. GitHub Repository

GitHub Link:
https://github.com/omkathukmvlu-hue/Data-Visualization-with-Power-BI-andTableau-Project.git
Recommended files to upload:

- Tableau workbook (.twb / .twbx)
- Dataset (.csv / .xlsx)
- Dashboard screenshot
- README file